JONATHAN WESTERFIELD

+1 713-253-6097 | jonathan.g.westerfield@gmail.com | LinkedIn | GitHub

PROFESSIONAL SUMMARY

Backend & Platform Engineer with 5+ years at Amazon building **Java/Python services**, **APIs**, and reliable **distributed systems**. Currently building Librarian, an open-source local-first retrieval platform, while pursuing backend and platform roles.

TECHNICAL SKILLS

Languages	Python, Java, Lua, Bash
AI & Search	RAG, Semantic Search, LLM Workflow Design, Local LLM Tooling
Backend	FastAPI, Docker, Docker Compose, Linux, Git, SQLite, Shell Scripting, Object-Oriented Programming
Cloud & Systems	AWS EC2, CloudWatch, Distributed Systems, Asynchronous APIs, CoAP, Middleware Design, API Contracts, CI/CD
AI Hardware	AWS Trainium, AWS Inferentia, HBM, BMC, PCIe, SoC Diagnostics

WORK EXPERIENCE

Amazon — Annapurna Labs 12/2022 – 07/2025
Software Development Engineer II — Machine Learning Acceleration Systems *Austin, TX*

- Unblocked debugging workflows across 8+ engineering teams by integrating **AWS CloudWatch telemetry** into fleet-wide testing infrastructure, reducing cross-organizational investigation time-to-resolution by over 90%.
- Designed and owned a **Lua/POSIX-based asynchronous CoAP API framework** with 100+ config-driven endpoints, improving test development speed by 30% and hardware diagnostic speed by over 55%.
- Led fleet-wide **root-cause analyses** using **AWS CloudWatch**, debuggers, and hands-on hardware analysis, reducing failure rates by over 60% per investigation.

Amazon — Amazon Advertising 05/2021 – 12/2022
Software Development Engineer — Amazon Ad Exchange (AAX) *Austin, TX*

- Served as primary technical owner for Amazon-owned advertising surfaces including Amazon Retail, Alexa, and Twitch, coordinating technical decisions across multiple partner organizations.
- Accelerated development and **production testing** for new pricing mechanics by over 85% by leading cross-team design of **Java/Spring-based auction experimentation controls**, enabling page & placement-level **A/B testing**.
- Implemented **Java-based auction ad deduplication** capabilities for a new pricing flow, protecting advertiser budgets, enabling adoption by partner teams, and safeguarding over $70MM in Amazon Retail revenue.

Amazon — Amazon Web Services 06/2020 – 05/2021
Software Development Engineer — AWS Directory Service *Seattle, WA*

- Remediated a critical security vulnerability by migrating 7+ code packages to dedicated **pre-production aliases**, improving **deployment isolation** and **release safety**.
- Led **Python 3 migration** of the primary **canary monitoring system** and established a **repeatable upgrade path**.

PERSONAL PROJECTS

Librarian — Open-Source AI Project 07/2025 – Present
Independent Software Engineer *Remote | GitHub*

- Building a **local-first RAG platform** using **FastAPI**, **Docker**, **SQLite**, and **local LLMs** to ask queries about long-form ebook files and get answers as if speaking to a librarian.
- Implemented **EPUB ingestion**, **book metadata extraction**, **retrieval**, and **source-grounded Q&A**.

EDUCATION

Texas A&M University: B.S. Computer Engineering *Graduated December 2019*